ϕ^4 in 4D : One loop $\beta(\lambda)$ and the running coupling.

Simon suggests that a good place to start may be the 0D analog, which is the integral

$$\int dx \, \exp\left(-\frac{x^2}{2} - g\frac{x^4}{4!}\right)$$

And then move to 1D (Quantum Mechanics) before moving to 4D (QFT)

The following is some notes mostly form online / AI sources to get some background.

> The dimension-by-dimension hierarchy (from 0D (calculus) to 1D (QM) to 4D (QFT) is the cleanest way to demystify QFT. At each step up in dimension the core mathematical structure remains identical, but the physical degrees of freedom expand, introducing the very divergences that require normalisation.

Dimension	Domain	Integration Var	Path Integral / Partition fn	Divergence
0D	$\mathbb{R}^1$	Single Var x	$Z(g) = \int dx\, e^{-S(x)}$	None.
1D	QM	Paths $x(t)$	$= \int Dx(t)\, e^{-S[x(t)]}$	None
4D	QFT	fields $\phi(x)$	$= \int D\phi(x)\, e^{-S[\phi(x)]}$	UV divergence $\int d^4x$

0D Pure calculus & Asymptotic series.

In zero spacetime dimensions, field theory reduces to standard single variable integration over a continuous variable x

$$Z(g) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx \, \exp\Big(\underbrace{-\tfrac{1}{2}x^2}_{\text{Free action}} \underbrace{- \tfrac{g}{4!}x^4}_{\text{Interaction term}}\Big)$$

$$\boxed{= \int e^{-S(x)} dx}$$

for small g we can expand the interaction exponent.

$$e^{-\frac{g}{4!}x^4} \simeq 1 - \frac{g}{4!}x^4 + \left(\frac{g}{4!}\right)^2 \frac{x^8}{2!} - \quad \dots$$

$$\simeq \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \left(\frac{g}{4!}\right)^n x^{4n}$$

$$\Rightarrow Z(g) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dx \, e^{-\frac{x^2}{2}} \sum \dots$$

we can evaluate $\int dx\, e^{-\frac{x^2}{2}} x^{4n} = \underbrace{\sqrt{2\pi}\,(4n-1)!!}$

* check this

note !! is the gamma function.

$$\Rightarrow Z(g) = \sum_{n=0}^{\infty} (-1)^n \frac{(4n)!}{2^{2n}(2n)!\, n!\, (4!)^n} g^n$$

$$= 1 - \frac{1}{8}g + \frac{35}{384}g^2 - \frac{385}{3072}g^3 + \dots \qquad ①$$

Note that $(4n-1)!!$ expressed via standard factorials is.

$$\frac{(4n)!}{4^n (2n)!}$$

Aside - The reason we have expressed this as a series expansion is two-fold

1) It gives us physical insight in that each term maps directly to a Feynman diagram
2) We can compute this integral exactly in 1D (in python / Mathematica) but in 4D this fails as it is an infinite dimensional integral.

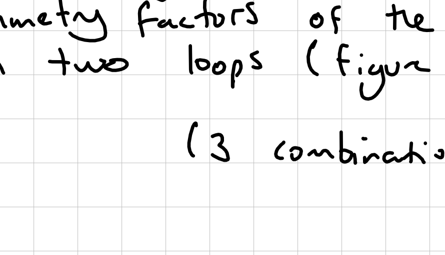
Feynman Diagrams

The terms in the expansion ① above can be represented by Feynman diagrams.

- Propagator $\langle x^2 \rangle_0 = 1$ ← first term in ① — a single line (why x^2?)
- Vertex $\frac{g}{4!}$ — a point joining for lines.
- The coefficient of g^1 (1/8) comes directly from counting symmetry factors of the 4-point vertex contracted in two loops (figure 8)

$$\frac{3}{4!} = \frac{1}{8} \qquad \text{(3 combinations)}.$$

0D 1-loop



i) Linear term (g^1) - second term in ①

The expansion gives $-\frac{g}{4}\langle x^4 \rangle_0$

- The denominator $4! = 24$ is the symmetry factor built into the definition of the vertex $\frac{g}{4!}x^4$

It prevents overcounting when you permute the identical fields attached to the vertex

- The three diagrams, $\langle x^4 \rangle_0 = 3$. By Wicks theorem computing the expectation value means pairing up the 4 legs in all possible ways. The number of ways to fully pair $2n$ items is $(2n+1)!!$.

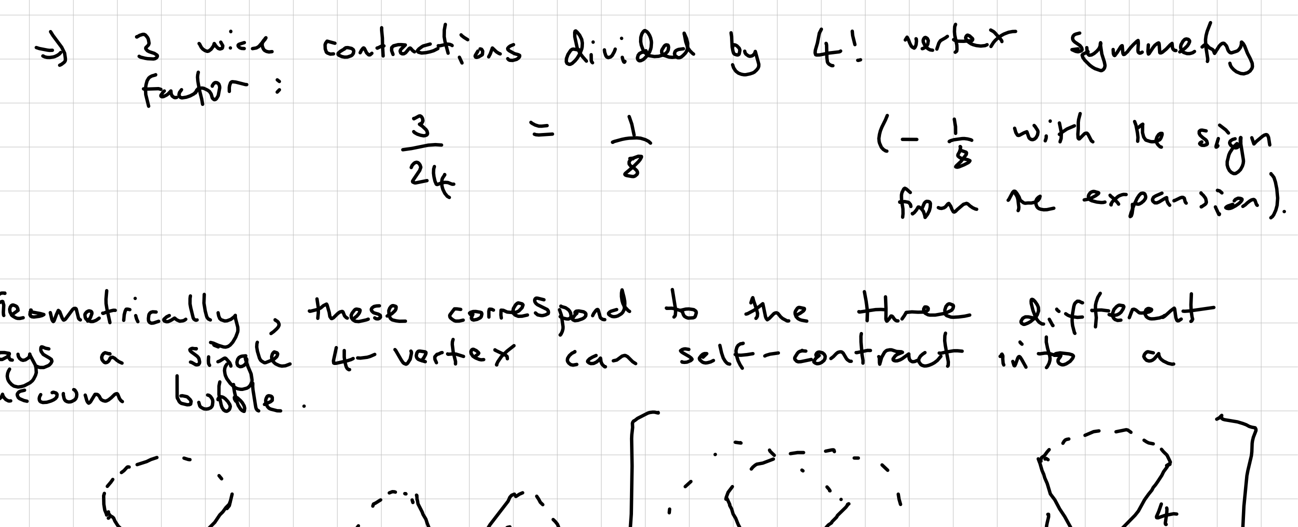
for four legs this is $(4-1)!! = 3 \times 1 = 3$

- $\Rightarrow$ 3 wick contractions divided by 4! vertex symmetry factor:

$$\frac{3}{24} = \frac{1}{8}$$

($-\frac{1}{8}$ with the sign from the expansion).

Geometrically, these correspond to the three different ways a single 4-vertex can self-contract into a vacuum bubble.

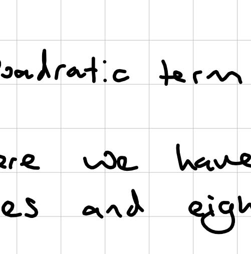


This is just one topological shape (single vertex with two self loops), but there are three ways to pair the four legs.

ii) Quadratic term g^2

Here we have three topologies. Note we have two vertices and eight legs

a)

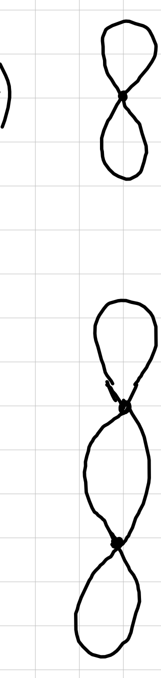


Double figure of eight

3 × 3 combinations = 9

[K = 0 no lines connecting vertices

b)

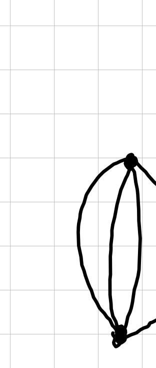


Dumbell diagram with two lines connecting vertices.

[K = 2 two lines connecting vertices

c)

Quadruple link (no self loops)



[K = 4

The number of combinations is $(2n-1)!! = (8-1)!!$

$= 105$

$\Rightarrow$ the coefficient is given by

$$\underbrace{\frac{1}{2!}}_{\text{Taylor expansion prefactor}} \times \underbrace{\left(\frac{1}{4!}\right)^2}_{\text{vertex denominator}} \times \underbrace{105}_{\text{Wick contractions}} = \frac{105}{1152} = \boxed{\frac{35}{384}}$$

Note that as we move to higher orders (g^3, g^4 etc.) the number of Wick contractions grows factorially

$$(2n-1)!!$$

so these series are asymptotic rather than convergent.